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Use of a Chelate Complex with Boron as a Lithium Salt for Lithium Battery Electrolytes

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ABSTRACT

A chelate complex with boron, lithium bis[2,3-naphthalenediolato(2-)-O,O']borate was used in lithium battery electrolytes for $\text{Li/V}_2\text{O}_5$ prototype cells. This lithium salt remains a safe and thermally stable compound up to 320°C without melting. A high specific conductivity (4.63 mS cm^{-1}) is exhibited in spite of its high viscosity (1.732 cP) in a 0.5 mol dm^{-3} electrolyte/ethylene carbonate-1,2-dimethoxyethane (mol ratio 2:3) binary solvent mixture at 25°C . Based on the discharge characteristics of a $\text{Li/V}_2\text{O}_5$ prototype cell, this lithium salt electrolyte resulted in high energy density (460 Wh kg^{-1}) at a cutoff potential of 2.5 V vs. Li/Li^+ .

Introduction

Some lithium salts based on anions with a high level of charge delocalization have been synthesized as lithium battery electrolytes. For example, the available lithium imide salt, $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ and methide salt, $\text{LiC}(\text{CF}_3\text{SO}_2)_3$ are reported to offer good characteristics: thermal and electrochemical stabilities and high conductivity in liquid and polymer solvent systems.¹⁻⁴ These lithium salts produce highly conductive solutions in low-dielectric ether-based solvent systems commonly used in secondary lithium batteries due to their highly charge-delocalized structures with the strong electron-withdrawing power of the terminal $-\text{CF}_3$ groups. They show useful conductivity not only in a liquid-solvent system but also in a polymer-solvent system. Recently, a new class of electrochemically and thermally stable lithium salt, which has a unique anion of a chelate complex with boron, has been described for new lithium battery electrolytes. The borate coordination compound so far has been applied as the solute in the electrolyte of electrolytic capacitors.^{5,6} Lithium bis[1,2-benzenediolato(2-)-O,O']borate (LBBB) and lithium bis[2,3-naphthalenediolato(2-)-O,O']borate (LBNB) have sparked considerable interest in their use as lithium battery electrolytes by Barthel *et al.*^{7,8} The bidentate ligands make it possible to achieve improved safety characteristics and higher thermal stabilities than monodentate ligands.⁷ The large anionic radius and charge delocalization due to aromatic substituents can be effective for its solubility in aprotic solvents and certain ethers. Furthermore, the conductivities and operating potential windows (limited by the oxidation of the borate) are modified by the presence of fluorinated anions with electron-withdrawing groups.^{8,9}

We report here on the good characteristics of LBNB as a lithium salt for lithium battery electrolytes in comparison with LBBB. Their thermal stabilities and conductivities in binary solvent mixtures are investigated, and the use of these electrolytes in $\text{Li/V}_2\text{O}_5$ prototype cells is examined.

Experimental

LBBB was synthesized and purified as described in the literature.⁷ LBNB was synthesized by modification of a literature procedure.⁸ Lithium hydroxide monohydrate (13.01 g , 0.31 mol) and boric acid (19.30 g , 0.31 mol) were dissolved in distilled water (0.6 dm^3) at 40°C in a round-bottom separation flask, and the solution was heated to near its boiling point. 2,3-Dihydroxynaphthalene

(100 g , 0.62 mol) was slowly added to the solution, which was then refluxed under an inert atmosphere for 30 min . After cooling the reaction mixture, a white solid precipitated immediately from the orange solution. The flask was maintained at room temperature for 24 h and then stored in a refrigerator (5°C) for 24 h . The product was filtered off, dried *in vacuo* at room temperature for 12 h and at 120°C for 48 h , and the yield was 80 g (77%). For purification of the compound, recrystallizations from acetone were repeated several times until the orange solution became colorless. The white solid compound was carefully dissolved in boiling acetone (0.35 dm^3), and stirred for 30 min under an inert atmosphere. After cooling to 5°C , a white crystalline compound was obtained, filtered off, and dried *in vacuo* as mentioned above. The purified compound was finally obtained after five recrystallizations, and the yield was 30 g (29%). The purity of the compound was confirmed by elemental analysis, using inductively coupled plasma emission spectrometry for both boron and lithium contents, and we found: C, 68.5 ; B, 3.24 ; Li, 1.98 ; and H, 3.60 (%). These data may be compared to calculated values: C, 71.91 ; B, 3.24 ; Li, 2.08 ; H, 3.62 (%). The structure of the compound was determined by $^{13}\text{C}\{^1\text{H}\}$ COSY (C-H shift correlation spectroscopy) nuclear magnetic resonance (NMR) spectrum (Fig. 1) in acetone- d_6 with 1% tetramethylsilane (TMS). This spectrum was recorded at room temperature on a JEOL JNM-EX400WB FT NMR system.

Conductivities of the electrolytes were measured using a conductometer (Toa Electronics Co., Model CM-60S) with a standard cell (Model CGT-511B). Simultaneous thermogravimetric (TG) and differential thermal (DTA) analyses were carried out with a Seiko Instruments Inc. TG/DTA220 system using a sample mass of about 10 mg . Discharge characteristics of the electrolytes were measured by a six-channel charge-discharge unit (Hokuto Denko Co., Model HJ-101SM6). An airtight prototype cell (Hohsen Co.) which has a V_2O_5 (Toshiba Battery Co., Ltd.) positive electrode, Li metal as the negative electrode, and a glass fiber filter (Advantec Toyo GA-100) as the separator, was used.

Results and Discussion

Figure 2 shows the simultaneous TG and DTA curves of LBNB and LBBB. LBNB begins to decompose at high temperature 320°C , and a large mass loss occurs at about 400°C with two exothermic peaks visible on the DTA. In LBBB, the decomposition begins at 250°C and a large mass loss occurs at about 350°C with an exothermic peak visible on the DTA. This result shows that the

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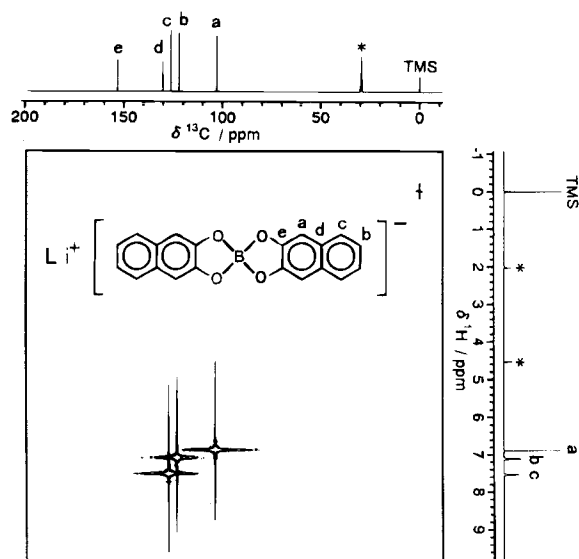


Fig. 1. C-H COSY NMR spectra of LBNB in acetone- d_6 with 1% TMS. The symbols in the spectrum were assigned to those of protons and carbons in LBNB. *: Peaks due to acetone- d_6 .

decomposition temperatures of LBNB are much higher than those of LBBB, and thermal stability is much improved by replacing benzene with naphthalene in the ligand portion of the anion. It is clear that the thermal stability of LBNB is superior to LBBB for application in lithium battery electrolytes. The difference in the decomposition process can be linked to the difference of the anion structure, that is, LBNB has a naphthalene moiety and LBBB has a benzene moiety in each anion. The high thermal stability of LBNB can be linked to the character of naphthalene moiety. This means that the conjugation energy of the aromatic ring system of naphthalene is larger than that of benzene. Finally at 500°C, LBNB and LBBB are shown to change into the nontoxic and safe compounds Li_2O and B_2O_3 .

Electrolytic conductivities of selected 0.5 mol dm^{-3} LBNB and LBBB solutions at 25°C are listed in Table I. Conductivities of both lithium salts are higher in binary solvent systems when compared with those of solutions based on propylene carbonate (PC) alone. The enhancement of conductivity on addition of DME is due to the lower viscosity of the solvent which renders ion migration more facile as mentioned in the literature.¹⁰⁻¹² The maximum conductivity of LBNB solution arises in an EC-DME (mol ratio 2:3) binary mixture. The value (4.63 $mS\ cm^{-1}$) is relatively high considering the high viscosity of the solution (1.732 cP) and the large molecular weight of LBNB (334.06), whereas LBBB solution exhibits 5.05 $mS\ cm^{-1}$ and 1.314 cP in this mixture. The result can be explained by the higher ionic dissociation ability of LBNB than that of LBBB.

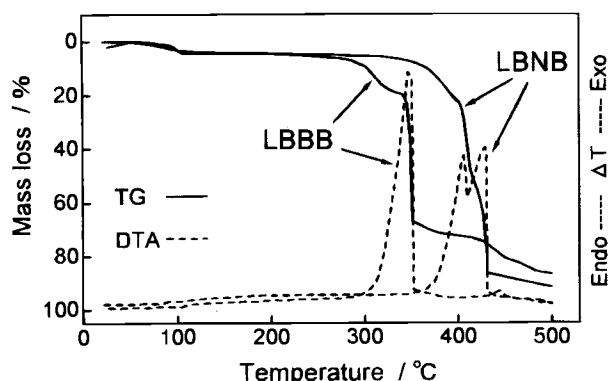


Fig. 2. Thermogravimetric and differential analysis curves of LBNB and LBBB at a heating rate of 1°C min^{-1} in air. α -Alumina was used as the reference.

Table I. Specific conductivities (κ) of 0.5 mol dm^{-3} LBNB and LBBB solutions at 25°C.

Solvent system (mol ratio)	κ ($mS\ cm^{-1}$)	
	LBNB	LBBB
PC	1.22	1.61
PC-DME (1:1)	3.84	4.03
PC-DME (3:7)	4.11	3.82
EC-DME (1:1)	4.51	5.04
EC-DME (2:3)	4.63	5.05
PC-EC (1:1)	1.39	1.75
PC-EC (3:7)	1.50	1.90

In LBNB, the coulombic force between lithium ion and anion should be weak because of the effect of the large charge delocalization in the large aromatic naphthalene diol anions. The characteristics of the large charge delocalization of the anion in LBNB were anticipated by the mean charge at the oxygen atoms from semiempirical quantum-mechanical calculations.⁸

Figure 3 shows the discharge characteristics of 0.5 mol dm^{-3} LBNB, LBBB, and other electrolytes in EC-DME (mol ratio 1:1) binary mixtures at 25°C. The discharge curve obtained in LBNB solution exhibits the usual two plateau potentials between 3.4 and 2.9 V vs. Li anode due to the Li^+ intercalation processes in $Li_{0.5}V_2O_5$ and LiV_2O_5 . The average discharge potential above 2.5 V in LBNB solution is 3.0 V and nearly equal to that for another common electrolyte, $LiBF_4$; 3.2 V, $LiBPh_4$; 3.1 V, $LiCF_3SO_3$; 3.1 V, $LiClO_4$; 3.0 V, and $LiPF_6$; 3.0 V. These average discharge potentials and capacities are slightly different from each other. This may be caused by the variable influence of an anion on the Li^+ intercalation process, based on the different interaction between an anion and the anode or cathode material in each electrolyte solution. A detailed surface study should be undertaken to investigate this phenomenon. The capacity of LBNB solution approaches the theoretical capacity (150 Ah kg^{-1}) of the cathode material above 2.5 V, and the LBNB solution system exhibits high energy density (460 Wh kg^{-1}) for a cutoff potential of 2.5 V vs. Li/Li^+ . LBNB may find application as a good lithium salt in lithium battery electrolytes as well as other common electrolytes. The discharge curve of the LBBB solution system drops early and exhibits a lower energy density (380 Wh kg^{-1}). This may be a result of the low oxidation potential (3.6 V)⁷ and relative instability of the cathode material V_2O_5 in LBBB solution. The LBBB solution shows some problems in that it becomes green by dipping the cathode into it for 2 h, and after discharge the solution becomes more deeply colored, whereas LBNB solution never exhibits such problems. Accordingly, LBNB is expected to be a good lithium salt for lithium battery electrolytes.

Conclusion

A new class of lithium salts as a chelate complex with boron, LBBB and LBNB were studied for possible application as lithium battery electrolytes. LBNB was thermally stable; it decomposed without melting above 320°C. The conductivities in aprotic binary-solvent (PC-DME, EC-DME, PC-EC) electrolytes were relatively high in

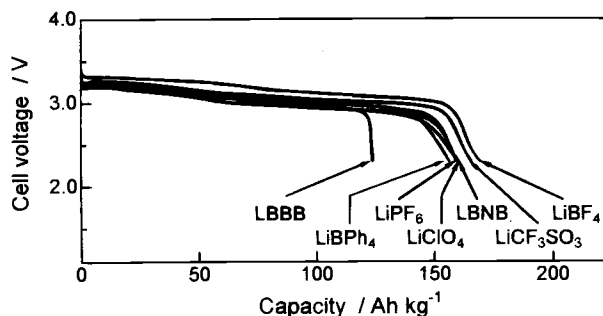


Fig. 3. Discharge curves in EC-DME (mol ratio 1:1) binary mixtures containing 0.5 mol dm^{-3} LBNB, LBBB, $LiBF_4$, $LiBPh_4$, $LiCF_3SO_3$, $LiClO_4$, and $LiPF_6$ for a Li/V_2O_5 cell at 25°C. Current density: 1.0 mA cm^{-2} .

spite of their high viscosities and the large molecular weight of the compounds. It became clear from the discharge characteristics of a $\text{Li}/\text{V}_2\text{O}_5$ prototype cell that this electrolyte had high energy density and electrochemical stability. These characteristics are a result of the electron-withdrawing properties of the large anion. Fluorinated compounds similar to these lithium salts are expected to show better electrochemical stability and a wide potential window range.

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REFERENCES

1. A. Webber, *This Journal*, **138**, 2586 (1991).
2. L. A. Dominey, V. R. Koch, and T. J. Blakley, *Electrochim. Acta.*, **37**, 1551 (1992).
3. C. W. Walker, Jr., J. D. Cox, and M. Salomon, *This Journal*, **143**, L80 (1996).
4. M. Ue, *ibid.*, **143**, L270 (1996).
5. R. W. Santway and R. S. Alwitt, U. S. Pat. 3,403,305 (1968).
6. M. Ue, K. Shima, and S. Mori, *Electrochim. Acta.*, **39**, 2751 (1994).
7. J. Barthel, M. Wühr, R. Buestrich, and H. J. Gores, *This Journal*, **142**, 2527 (1995).
8. J. Barthel, R. Buestrich, E. Carl, and H. J. Gores, *ibid.*, **143**, 3572 (1996).
9. J. Barthel, R. Buestrich, E. Carl, and H. J. Gores, *ibid.*, **143**, 3565 (1996).
10. S. Tobishima and A. Yamaji, *Electrochim. Acta.*, **28**, 1067 (1983).
11. S. Tobishima, J. Yamaki, and T. Okada, *ibid.*, **29**, 1471 (1984).
12. S. Tobishima and T. Okada, *ibid.*, **30**, 1715 (1985).

Performance Improvement of a PEMFC Using Fuels with CO by Addition of Oxygen-Evolving Compounds

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ABSTRACT

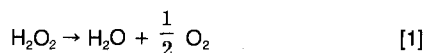
A new method is described to improve the performance of a proton exchange membrane fuel cell (PEMFC) using reformed methanol or H_2/CO as fuels. The addition of liquid hydrogen peroxide to the humidification water for the fuel gas leads to a heterogeneous decomposition of H_2O_2 and formation of active oxygen. In this way adsorbed CO is oxidized nonelectrochemically to CO_2 and the blocking of the hydrogen oxidation reaction at the anode can be avoided. It is demonstrated that a complete recovery of the CO-free performance is achieved for $\text{H}_2/100$ ppm CO.

Introduction

In a PEMFC operating at 80°C the performance of the cell drops significantly when hydrogen as the fuel gas contains CO¹ which is the case of reformed methanol or natural gas. This is due to the strong adsorption of CO on the Pt anode of the cell. Besides optimization of gas purification in the periphery of the fuel cell, several approaches are being discussed to solve the CO problem with electrochemical engineering. The use of PtRu alloys as anode catalyst has shown that CO is oxidized at more negative potentials as compared to pure Pt and that a significant improvement of the cell performance has been described for CO levels up to 250 ppm at cell temperatures of 80°C .^{2,3} However, the same current-voltage plot as in pure hydrogen could not be attained.

Another method has been described by Gottesfeld and Pafford consisting in the feed of oxygen into the fuel gas stream.⁴ Adsorbed CO is oxidatively removed. The same current-voltage characteristic as in pure hydrogen has been observed for 100 ppm CO in H_2 when 4.5% oxygen was injected.

Another approach may be the use of compounds that evolve oxygen under operating conditions of the fuel cell. An oxygen carrier may be hydrogen peroxide that decomposes according to the following reaction



It is possible to store liquid hydrogen peroxide⁵ separately in a tank and to add it to the humidification water for the fuel gas. The advantage of such indirect oxygen bleeding is evident regarding the safety problems with respect to handling hydrogen/oxygen gas mixtures at 80°C in a fuel cell having highly active noble metal catalysts for the nonelectrochemical recombination of H_2 and O_2 . $\text{H}_2\text{O}_2/\text{H}_2\text{O}$ mixtures are relatively safe. Explosive vapor mixtures

are formed only when the H_2O_2 concentration in the vapor phase exceeds 26 mole percent (m/o, at atmospheric pressure).⁵

The first results on cell performance of a PEMFC in $\text{H}_2/100$ ppm CO humidified with $\text{H}_2\text{O}_2/\text{H}_2\text{O}$ mixtures are reported in this communication.

Experimental

Membrane electrode assemblies (MEAs) with a catalyst loading of $1 \text{ mg}/\text{cm}^2$ were made as described in Ref. 3. Gas diffusion electrodes from E-TEK (Natick, MA) were used with Pt as cathode and $\text{Pt}_{0.7}\text{Ru}_{0.3}$ as anode catalyst. The MEA was mounted in a laboratory test cell with Ti current collectors and gas manifolds. The active electrode area was 20 cm^2 . The cell performance was evaluated in a test station with mass flow, temperature, and pressure control, and an electronic load (HP 6060B) for recording current-voltage curves. Pure hydrogen and $\text{H}_2/100$ ppm CO (Messer-Griesheim) were used as fuel gases. At the outlet of the anode a constant gas flow of $17 \text{ ml}/\text{min}$ was used independent of the applied current density. Pure oxygen with a constant excess of $30 \text{ ml}/\text{min}$ was used at the cathode. The cell temperature was 80°C and the pressures in the anode and cathode compartments were 0.22 and 0.25 MPa, respectively. The humidification of the fuel gas was carried out by injecting the dry fuel gas into a flowing vapor stream at 100°C . The vapor/fuel gas stream was then passed through a condenser at 40°C thus adjusting the water partial pressure. For CO removal the humidification water contains 1 to 5 volume percent (v/o) hydrogen peroxide (Merck, 30%, stabilized).

Results and Discussion

Current-voltage curves were measured in pure hydrogen and oxygen without H_2O_2 addition to the humidification water. Under operating conditions a current density of about $0.38 \text{ A}/\text{cm}^2$ was obtained at a cell voltage $U_{\text{cell}} = 0.7 \text{ V}$ (Fig. 1). When the fuel gas was changed from H_2 to $\text{H}_2/100$ ppm CO at constant current density, the cell voltage decreased continuously until a steady state was attained.³ Figure 1 shows the resulting steady-state current-voltage plot in $\text{H}_2/100$ ppm CO. Under this condition the current

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